EXERCISES ON LIMITS & COLIMITS

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Exercise 1. Prove that pullbacks of epimorphisms in **Set** are epimorphisms and pushouts of monomorphisms in **Set** are monomorphisms. Note that these statements cannot be deduced from each other using duality. Now conclude that the same statements hold in **Top**.

Exercise 2. Let X be a set and $A, B \subset X$. Prove that the square

$$\begin{tikzcd} A \cap B \arrow[r, hook] \arrow[d, hook] & A \arrow[d, hook] \\ B \arrow[r, hook] & A \cup B \end{tikzcd}$$

is both a pullback and pushout in **Set**.

Exercise 3. Let X be a space. Prove that the *opposite* of the poset $\mathrm{Op}(X)$ of open subsets of X is filtered.

Exercise 4. Let X be a space and $x \in X$. Write $\mathrm{Op}_x(X)$ for the full subposet of $\mathrm{Op}(X)$ spanned by those open subsets containing the point x. Prove that $\mathrm{Op}_x(X)^{op}$ is filtered.

Exercise 5. Prove that the subcategory of **CRing** of domains is ***stable under filtered colimits***, i.e., that if $F\colon I \longrightarrow \mathbf{CRing}$ be a filtered diagram of commutative rings, where $F(i)$ is a domain for all $i \in I$, then $\mathrm{colim}_I F$ is a domain (where the colimit is taken in **CRing**).

You'll need to use the explicit description of filtered colimits in **Set**, which also describes the underlying set of the filtered colimit in **CRing**. You should think about what the ring structure should be. More generally, this description gives the underlying set of a filtered colimit in $\mathbf{Mod}_R$ or $\mathbf{Alg}_R$, where R is a commutative ring (taking $R = \mathbf{Z}$, we recover the special cases of $\mathbf{Ab} = \mathbf{Mod}_{\mathbf{Z}}$ and $\mathbf{CRing} = \mathbf{Alg}_{\mathbf{Z}}$).

Exercise 6. Let C be a category and $f\colon X \longrightarrow Y$ be a morphism in C. Prove f is a monomorphism if and only if the square

$$\begin{tikzcd} X \arrow[r, equal] \arrow[d, equal] & X \arrow[d, "f"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

is a pullback square.

Exercise 7. Suppose that C and D are categories and $F, G\colon C \longrightarrow D$ functors. Prove that if $\alpha\colon F \Longrightarrow G$ is a natural transformation whose components are monomorphisms in D, then α is a monomorphism in D^C. Conversely, if C is small, D has pullbacks, and α is a monomorphism in D^C, prove that each component of α is a monomorphism in D.

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Definition. Let $\mathcal{F}$ be a presheaf of sets (or commutative rings, or abelian groups) on a space X. The ***stalk*** of $\mathcal{F}$ at x is the colimit

$$\mathcal{F}_x \coloneqq \operatorname*{colim}_{\mathrm{Op}_x(X)^{op}} \mathcal{F}$$

(of the diagram $\mathcal{F}$ restricted to $\mathrm{Op}_x(X)^{op}$).

Exercise 8. Let $\phi\colon \mathcal{F} \rightarrowtail \mathcal{G}$ be a monomorphism of presheaves of sets on a space X. Prove that for all $x \in X$, the induced map on stalks $\phi_x\colon \mathcal{F}_x \longrightarrow \mathcal{G}_x$ is an injection.

Remark. As it turns out, the converse to Exercise 8 is also true, assuming that $\mathcal{F}$ is a sheaf, but it's not as elementary.

Definition. Let $\mathcal{F}$ be a presheaf of sets (or commutative rings, or abelian groups) on a space X, $f \in \mathcal{F}(X)$, and $x \in X$. Write f_x for the image of f in the stalk $\mathcal{F}_x$ under the leg $\lambda_X\colon \mathcal{F}(X) \longrightarrow \mathcal{F}_x$ of the colimit cone. The element f_x is called the ***germ*** of f at x.

Exercise 9. Let $\mathcal{F}$ be a presheaf of commutative rings on a space X, and $f \in \mathcal{F}(X)$. Prove that the set

$$\{\, x \in X \mid f_x \in \mathcal{F}_x^\times \,\} \subset X$$

is open, where $\mathcal{F}_x^\times$ is the group of units of the stalk $\mathcal{F}_x$.

Exercise 10. Let R be a commutative ring. Prove that every R-module can be written as a filtered colimit of its finitely generated submodules.

Definition. Let X be a space and S a set. The ***constant presheaf on*** X ***with value*** S is the constant functor $S\colon \mathrm{Op}(X)^{op} \longrightarrow \mathbf{Set}$ at S.

Exercise 11. Let X be a space. For which sets S is the constant presheaf at S a sheaf?
Hint: The axioms for a topology on a set X guarantee that certain subsets of X are open, and the value of the constant presheaf is completely determined by where it sends any open subset $U \subset X$.

Exercise 12. Let X be a set. Give a categorical definition of a topology on X as a subposet of the power set of X (regarded as a poset under inclusion) that is stable under certain categorical constructions.

Exercise 13. Let X be a space. Give a categorical description of what it means for a set of open subsets of X to form a basis for the topology on X.

Lemma. *Let C be a category. If the identity functor $\mathrm{id}_C\colon C \longrightarrow C$ has a limit, then $\lim_C \mathrm{id}_C$ is an initial object of C.*

Proof. To simplify notation, write $\varnothing \coloneqq \lim_C \mathrm{id}_C$. Let $\lambda\colon \varnothing \Longrightarrow \mathrm{id}_C$ be the limit cone. Since $\varnothing$ admits a morphism λ_c to every object $c \in C$, it suffices to show that if $f\colon \varnothing \longrightarrow c$ is a morphism in C, then $f = \lambda_c$.

Since λ is a cone, the triangle

$$\begin{tikzcd}
 & \varnothing \arrow[dl, "\lambda_\varnothing"'] \arrow[dr, "\lambda_c"] & \\
\varnothing \arrow[rr, "f"'] & & c
\end{tikzcd}$$

commutes, hence it suffices to show that $\lambda_\varnothing = \mathrm{id}_\varnothing$. To see this, notice that since λ is a cone, for all $c \in C$, the triangle

$$\begin{tikzcd} & \varnothing \arrow[dl, "\lambda_\varnothing"'] \arrow[dr, "\lambda_c"] & \\ \varnothing \arrow[rr, "\lambda_c"'] & & c \end{tikzcd}$$

commutes. However, we also know that the triangle

$$\begin{tikzcd} & \varnothing \arrow[dl, equal] \arrow[dr, "\lambda_c"] & \\ \varnothing \arrow[rr, "\lambda_c"'] & & c \end{tikzcd}$$

commutes for all $c \in C$. Since λ is a limit cone, we have that $\lambda_\varnothing = \mathrm{id}_\varnothing$, as desired. □